9th Class • Punjab Board • PECTAA New Syllabus

Chapter 3: Introduction to Computer Networks

Complete solved exercise — protocols, OSI, topologies, security and worked transmission-time numericals.

Chapter 3 • Complete official exercise • 9 MCQs + 10 short + 6 long questions

What this chapter is about

• A **computer network** links devices so they can exchange data and share resources. Data communication needs a sender, receiver, message, medium and agreed protocols.

• Routers connect networks; switches connect devices within a local network. DNS resolves names, DHCP assigns configuration and TCP/UDP carry application data in different ways.

• Network **topology** describes device layout. The **OSI model** divides network communication into seven understandable layers.

• Simplex, half-duplex and full-duplex describe direction; circuit and packet switching describe how a communication path is used. Firewalls and encryption help protect data.

Coverage: Chapter 3 corresponds to Unit 6 Introduction to Computer Networks in the twelve-unit textbook. The April 2026 nine-unit exercise is solved below in its original order.

A. Multiple Choice Questions

1. The primary objective of computer networks is:

(a) Increase computational power
(b) Enable resource sharing and data communication
(c) Enhance graphic capabilities
(d) Improve software development

Answer: (b) Enable resource sharing and data communication — Networks allow users to share files, printers and Internet access and to exchange data efficiently.

2. Which device is used to connect multiple networks and direct data packets between them?

(a) Switch
(b) Hub
(c) Router
(d) Modem

Answer: (c) Router — A router forwards packets between different networks using network-layer addresses and routing information.

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3. Which layer of the OSI model is responsible for node-to-node data transfer and error detection?

(a) Physical Layer
(b) Data Link Layer
(c) Network Layer
(d) Transport Layer

Answer: (b) Data Link Layer — The Data Link layer frames data for delivery across a local link and detects transmission errors.

4. The function of the Domain Name System (DNS) is to:

(a) Assign IP addresses dynamically
(b) Translate domain names to IP addresses
(c) Secure data communication
(d) Monitor network traffic

Answer: (b) Translate domain names to IP addresses — DNS resolves human-readable names such as example.com to IP addresses used for routing.

5. Which method of data transmission uses a dedicated communication path?

(a) Packet Switching
(b) Circuit Switching
(c) Full-Duplex
(d) Half-Duplex

Answer: (b) Circuit Switching — Circuit switching establishes a dedicated path for the communication session; packet switching does not reserve one fixed path.

6. Which protocol is used for reliable data transfer in the TCP/IP model?

(a) HTTP
(b) FTP
(c) TCP
(d) UDP

Answer: (c) TCP — TCP provides connection-oriented delivery with acknowledgements, retransmission and ordered data.

7. The main purpose of a firewall in network security is to:

(a) Convert data into a secure-format
(b) Monitor and control network traffic
(c) Assign IP addresses
(d) Translate domain names

Answer: (b) Monitor and control network traffic — A firewall permits or blocks incoming and outgoing traffic according to security rules.

8. Which network topology connects all devices to a central hub?

(a) Ring
(b) Mesh
(c) Bus
(d) Star

Answer: (d) Star — In a star, each device has its own link to a central hub or switch.

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9. What is a key benefit of using computer networks in businesses?

(a) Increase computational power
(b) Enable resource sharing and efficient communication
(c) Enhance graphic capabilities
(d) Improve software development

Answer: (b) Enable resource sharing and efficient communication — Shared resources and fast communication reduce cost and support collaboration.

B. Short Questions

1. Define data communication.

Data communication is the exchange of digital data between two or more devices through a transmission medium according to agreed rules. It allows one device to send a message that another device can receive and understand.

2. Enlist its key components of data communications.

The five key components are the **sender**, **receiver**, **message** (the data), **transmission medium** (cable or wireless path) and **protocol** (the rules governing representation, delivery and timing).

3. Write the role of routers in a computer network.

A router connects different networks and forwards data packets towards their destination using IP addresses and routing information. It selects a suitable path, so it can link a home or school LAN to the Internet.

4. What are the main functions of the Network Layer in the OSI model?

The Network layer provides logical addressing and routes packets from source to destination across interconnected networks. It selects paths, forwards packets and may divide and reassemble packets when different networks require different sizes.

5. What is the purpose of the Dynamic Host Configuration Protocol (DHCP)?

DHCP automatically gives a device network configuration, normally an IP address, subnet mask, default gateway and DNS server information, for a defined lease period. This reduces manual configuration and addressing mistakes.

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6. Differentiate between TCP and UDP.

TCP	UDP
Connection-oriented; establishes a connection before data transfer.	Connectionless; sends datagrams without first establishing a connection.
Reliable: acknowledges data and retransmits lost segments.	Best-effort delivery with no built-in guarantee of arrival or retransmission.
Preserves order and provides flow and congestion control.	Does not itself guarantee order or provide TCP-style flow control.
Usually more overhead and delay.	Smaller header and lower protocol overhead; useful when speed matters.
Examples: web browsing, email and file transfer.	Examples: live audio/video, online games and DNS queries.

7. Explain the importance of encryption in network security.

Encryption converts readable plaintext into ciphertext using an algorithm and key. Unauthorised people who intercept the communication cannot easily understand it, so encryption protects confidentiality for messages, passwords, payments and stored data. It should be combined with authentication and integrity checks.

8. Define star topology?

Star topology is a layout in which every device has a separate connection to a central hub or switch. Failure of one device's cable normally affects only that device, but failure of the central device affects the whole network.

9. How does firewall contribute to network security?

A firewall monitors incoming and outgoing traffic and applies rules based on addresses, ports, protocols or application behaviour. It blocks unauthorised or suspicious connections while allowing approved traffic, forming a barrier between trusted and untrusted networks.

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10. What is difference between Half Duplex and Full Duplex communication?

Half-duplex	Full-duplex
Data can travel in both directions, but only one direction at a time.	Data can travel in both directions at the same time.
The two ends take turns to transmit.	Both ends may transmit and receive simultaneously.
Example: a walkie-talkie conversation.	Example: an ordinary telephone conversation.

C. Long Questions

1. Discuss the objectives of computer networks?

A computer network is created to make information and hardware available to connected users efficiently and reliably.

• **Resource sharing:** Users can share printers, scanners, storage, applications and an Internet connection instead of buying separate resources for every computer.

• **Data communication:** Files, messages, audio and video can move quickly between local or distant users.

• **Collaboration:** Teams can work on shared documents and communicate in real time even when they are in different places.

• **Central data management:** Shared servers and databases keep common records available and make backup, permissions and administration easier.

• **Cost saving:** Shared equipment, central maintenance and digital communication reduce hardware, travel and paper costs.

• **Reliability and availability:** Copies of data and alternative communication paths can keep services available if one component fails.

• **Scalability and remote access:** New users and devices can join the network, and authorised people can reach services from home or another city.

In a school, for example, one networked printer can serve a computer laboratory while a shared folder gives every student access to the same lesson files.

2. In a Simplex communication system, assume data is transmitted at a rate of 500 bits per second (bps). Compute the time to transmit a message if:
(a) it is of 10 kilobits. (b) it is of 10 kilobytes.

Simplex describes one-way transmission; it does not change the time formula. Use **time = message size in bits ÷ transmission rate in bits per second**.

(a) Message size = 10 kilobits

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In data-communication rates, the standard decimal convention is 1 kilobit = 1000 bits.

Message size = 10 × 1000 = 10,000 bits

Time = 10,000 bits ÷ 500 bits/s

Time = 20 seconds

(b) Message size = 10 kilobytes

Use the textbook's binary storage convention, 1 KB = 1024 bytes, and 1 byte = 8 bits.

Message size = 10 × 1024 bytes = 10,240 bytes

Message size in bits = 10,240 × 8 = 81,920 bits

Time = 81,920 bits ÷ 500 bits/s

Time = 163.84 seconds = 2 minutes 43.84 seconds

Convention check

• If a class uses 1 kilobit = 1024 bits, part (a) becomes 10240 ÷ 500 = **20.48 s**. If decimal kilobytes (1000 bytes) are used, part (b) becomes 80000 ÷ 500 = **160 s**.
• The main answers use **decimal kilobits** for transmission and the textbook's **1024-byte kilobyte** for storage. Always state the convention before calculating.

3. What is meant by protocols in data communication. Explain its different types.

A **protocol** is an agreed set of rules for communication. It defines matters such as data format, addressing, timing, error handling and the meaning of messages, allowing hardware and software from different producers to interoperate.

Protocol	Purpose
IP	Addresses and routes packets between networks; IPv4 and IPv6 are the main versions.
TCP	Provides reliable, ordered, connection-oriented delivery with retransmission and flow control.
UDP	Provides fast connectionless datagrams without guaranteeing arrival or order.
HTTP / HTTPS	Transfers web resources; HTTPS protects the HTTP exchange with encryption and server authentication.

Protocol	Purpose
FTP	Transfers files between a client and a server.
SMTP, POP3, IMAP	SMTP sends mail; POP3 and IMAP allow users to retrieve/manage mail from a server.
DNS	Resolves domain names to IP addresses and stores other naming records.
DHCP	Automatically supplies IP configuration to devices joining a network.

Protocols work together rather than alone. For example, opening a secure website may use DNS to find the server, IP to route packets, TCP for reliable delivery and HTTPS for an authenticated, encrypted web exchange.

4. Describe real-world applications of computer networks in business, education.

Business applications

• Shared files, printers, accounting systems and customer databases reduce duplicated equipment and keep business records consistent.

• Email, instant messaging and video meetings support quick communication among branches, suppliers and customers.

• E-commerce websites, online banking and digital payments let businesses sell and receive payments beyond a physical shop.

• Cloud services provide remote storage, backup and software; central monitoring and security help administrators protect systems.

• Point-of-sale and inventory networks update stock immediately, while analytics help managers make decisions from current data.

Educational applications

• Computer laboratories can share Internet access, printers, educational software and lesson files.

• Learning-management systems distribute lectures, assignments, grades and feedback; students can submit work remotely.

• Video classes and webinars connect teachers with learners in distant areas, while discussion groups support collaboration.

• Digital libraries and online research provide far more material than a single local collection.

• School-management networks maintain attendance, timetables, results and communication with parents centrally.

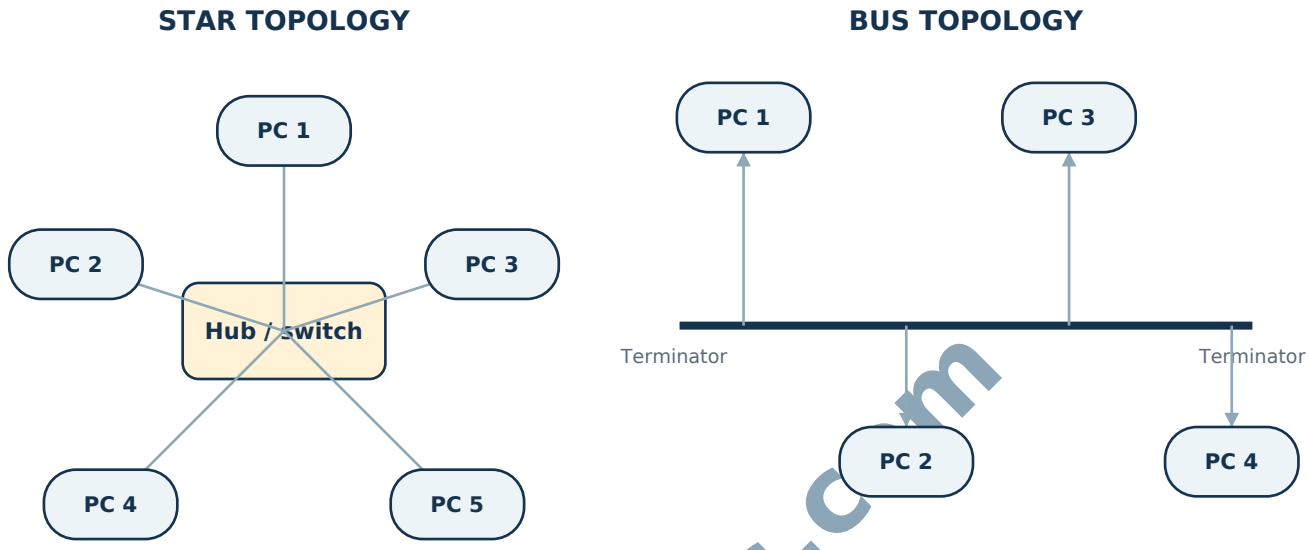
In both sectors, access rights, backups and safe online behaviour are necessary because shared access also creates security and privacy risks.

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5. What is network topology? Explain any two with diagrams.

A **network topology** is the physical or logical arrangement of devices and communication links in a network. Physical topology describes the actual layout; logical topology describes how data flows.

Figure 1 — Star and bus topologies



Every device uses a central device. All devices share one backbone cable.

Star topology

Each device has a separate link to a central hub or switch. It is easy to add a device and isolate a faulty cable, and one end device normally does not disturb the others. However, it needs more cable than bus and the central device is a single point of failure.

Bus topology

All devices share one backbone cable with terminators at both ends. It is simple and uses little cable, so it can be inexpensive for a small network. A backbone fault can disable the whole network, fault finding can be difficult and performance falls as traffic and device count grow.

Feature	Star	Bus
Connection	Separate link to central hub/switch	Shared backbone cable
Main advantage	Easy expansion and isolation of a device fault	Simple layout and less cabling
Main limitation	Central device failure affects all devices	Backbone failure affects all devices

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6. Define OSI network model, explain its any three layers.

The **Open Systems Interconnection (OSI) model** is a seven-layer conceptual framework that explains how data moves from an application on one device to an application on another. Each layer provides services to the layer above and uses the layer below.

No. / Layer	Main function	Example
7 — Application	Provides network services directly to user applications.	HTTP, DNS, SMTP
6 — Presentation	Formats, translates, compresses and encrypts data so applications can understand it.	Character/data encoding, TLS-related representation
5 — Session	Establishes, manages and ends communication sessions and dialogues.	Session setup and recovery
4 — Transport	Provides end-to-end delivery, segmentation, reliability and flow control as required.	TCP, UDP
3 — Network	Logical addressing and routing of packets between networks.	IP, routers
2 — Data Link	Frames data for a local link, handles MAC addressing and detects link errors.	Ethernet, switches
1 — Physical	Transmits raw bits through electrical, optical or radio signals and defines connectors/media.	Cables, radio signals, hubs

For the required three layers: **Physical** carries bits, **Data Link** delivers frames across a local link with error detection, and **Network** routes packets between networks using logical addresses. The remaining layers complete end-to-end application communication.

D. More Important Questions (from the text)

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1. Differentiate between LAN, MAN and WAN.

LAN	MAN	WAN
Local Area Network covers a home, room, school or office building.	Metropolitan Area Network covers a city or very large campus.	Wide Area Network covers countries or continents.
Usually owned and managed by one person or organisation.	Links several LANs across a metropolitan area.	Connects many LANs and MANs using service-provider links.
Normally high speed over short distances.	Medium geographical reach between LAN and WAN.	Long-distance communication; the Internet is the largest example.

2. Compare circuit switching with packet switching, and simplex with half-/full-duplex.

Circuit switching	Packet switching
A dedicated path is established for the whole session.	Data is divided into packets that may share links and follow available routes.
Resources remain reserved even during silence.	Network capacity is shared and used when packets are ready.
Delay is steady after setup, but setup takes time and a path failure interrupts the call.	Uses capacity efficiently; packets can experience queuing, loss or different delays.
Traditional telephone-call model.	Internet data communication.

Mode	Direction	Example
Simplex	One direction only.	Keyboard sending input to a computer (basic model).
Half-duplex	Both directions, one at a time.	Walkie-talkie.
Full-duplex	Both directions simultaneously.	Telephone conversation.

3. State five practices that improve network security.

• Use strong, unique passwords and multi-factor authentication for network and cloud accounts.

• Keep operating systems, routers, browsers and applications updated to close known vulnerabilities.

• Use a correctly configured firewall and secure Wi-Fi encryption; change default router passwords.

• Encrypt sensitive communication and verify websites/senders before providing information.

• Give users only necessary permissions, maintain tested backups and monitor logs for unusual activity.

Points to Remember

Data communication: sender + receiver + message + medium + protocol.

Router: forwards packets between networks; **switch:** connects devices within a LAN; **DNS:** name → IP; **DHCP:** automatic configuration.

TCP: reliable and ordered. **UDP:** connectionless and lower-overhead, without guaranteed delivery.

Topology: star uses a central device; bus uses one shared backbone. Each has a different single point of failure.

Direction: simplex = one way; half-duplex = both ways in turns; full-duplex = both ways simultaneously.

OSI layers: Physical, Data Link, Network, Transport, Session, Presentation, Application.

Transmission time: message size in bits ÷ rate in bps. Convert bytes to bits before dividing.

Security: firewall filters traffic; encryption protects confidentiality; updates, authentication and backups are also needed.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	b	2	c	3	b	4	b	5	b
6	c	7	b	8	d	9	b		

How to attempt this chapter in the paper

• For the numerical, write the size conversion, bit conversion and division separately, then attach seconds to the answer.

• In a topology answer, draw labelled devices and links and include at least one advantage and one limitation.

• For OSI answers, mention the layer number, its main job and a protocol/device example rather than listing names only.

• For protocol comparisons, use the exact qualities asked—reliability, order, connection, overhead and an application example.

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